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EDUCATION

Mar. 2018 ~ Feb. 2024	Sogang University Department of Computer Science and Engineering <i>B.S. in Computer Science and Engineering</i> GPA: 3.73 / 4.5	Seoul, Korea
Mar. 2024 ~ Feb. 2026	Sogang University Department of Computer Science and Engineering Thesis: Development and Performance Profiling of a Mobile Ray Tracing System Supporting Global Illumination Advisor: Insung Ihm <i>M.S. in Computer Science and Engineering</i> GPA: 4.32 / 4.5	Seoul, Korea

RESEARCH INTEREST

- Real-time rendering
- Mobile rendering
- Global illumination
- Ray tracing
- Physically Based Rendering (PBR)
- Game engine

PUBLICATIONS (INTERNATIONAL)

1. Taekgeun You, Woong Seo, Donghee Han, Insung Ihm, "Mobile-DDGI: Lightweight Probe-Based Global Illumination via Adaptive Budget Allocation", *I3D COMPANION '26: COMPANION PROCEEDINGS OF THE SYMPOSIUM ON INTERACTIVE 3D GRAPHICS AND GAMES*, (2026)

PUBLICATIONS (DOMESTIC)

1. Taekgeun You, Sungho Yun, Sehee Cho, Woong Seo, Insung Ihm, "Performance Profiling of Vulkan-based Ray Tracing Renderers on Mobile Platforms", *Journal of the Korea Computer Graphics Society*, (2025)

CONFERENCE

1. Taekgeun You, Sungho Yun, Sehee Cho, Woong Seo, Insung Ihm, "Performance Profiling of Vulkan-based Ray Tracing Renderers on Mobile Platforms", Korea Computer Graphics Society Conference 2025 (KCGS 2025), Goseong, Korea (Jul. 2025) - Conference paper
2. Taekgeun You, Woong Seo, Donghee Han, Insung Ihm, "Mobile-DDGI: Lightweight Probe-Based Global Illumination via Mobile-DDGI: Lightweight Probe-Based Global Illumination via", 026 ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D 2026), San Francisco, United States (May. 2026) - Poster
3. Taekgeun You, Woong Seo, Donghee Han, Insung Ihm, "Mobile-DDGI: Lightweight Probe-Based Global Illumination via Adaptive Budget Allocation", Korea Computer Graphics Society Conference 2026 (KCGS 2026), Yeosu, Korea (Jul. 2026) - Oral

RESEARCH EXPERIENCES

Sogang Computer Graphics Lab | Sogang University, Republic of Korea

Mar. 2024 ~ Feb. 2026

Graduate Research Assistant

Advisor: Insung Ihm

- Designed a mobile-optimized ray tracing renderer for real-time graphics applications.
- Developed hybrid ray tracing renderer based on deferred rendering and shadow map.
- Developed renderers based on ray queries and ray tracing extensions.
- Conducted various experiments to analyze rendering performance, image quality, and scalability across different scenes and sampling settings.

AWARD

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1. Best poster Award, ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D) (2026)

SELECTED PROJECTS

Vulkan-CUDA Selective Ray Tracer | *Independent Project* | C++, Vulkan, CUDA | Sep. 2024 – Dec. 2024

- Applied CUDA-based Sobel filtering to Vulkan shadow maps and traced rays only near detected shadow boundaries to reduce ray-tracing work while refining shadow edges.
- Implemented Vulkan-CUDA interoperability, external-memory sharing, and semaphore-based synchronization for cross-API GPU processing.

OpenGL Deferred Renderer | *Independent Project* | C++, OpenGL | Dec. 2024

- Built forward and deferred rendering pipelines with PBR shading and glTF scene loading.
- Optimized multi-light rendering using light-volume culling, depth/stencil tests, and selective pixel shading.

OptiX Ray Tracer | *Independent Project* | C++, OptiX | Dec. 2024

- Implemented reflection, refraction, shadows, PBR shading, instancing, glTF mesh rendering, and acceleration-structure updates for dynamic objects.
- Added custom intersection programs for cones and cylinders in addition to OptiX built-in primitives.

Mobile Gaussian Ray Tracer using Vulkan | *Team Project* | C++, Vulkan | May. 2025 – Jun. 2025

- Ported NVIDIA 3DGRUT to a Vulkan-based mobile renderer and implemented both ray-tracing-pipeline and ray-query variants for performance comparison.
- Optimized ray-query execution through BVH partitioning, shared-memory sorting, host-side command recording, and selective tracing of pixels with remaining transmittance.

SKILLS AND TECHNIQUE

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- Programming Languages: C, C++, C#, Python, glsl
 - APIs: OpenGL, Vulkan, OptiX, CUDA
 - IDE: Visual Studio
 - Tools: Git, Nsight, Radeon GPU Profiler/Raytracing Analyzer